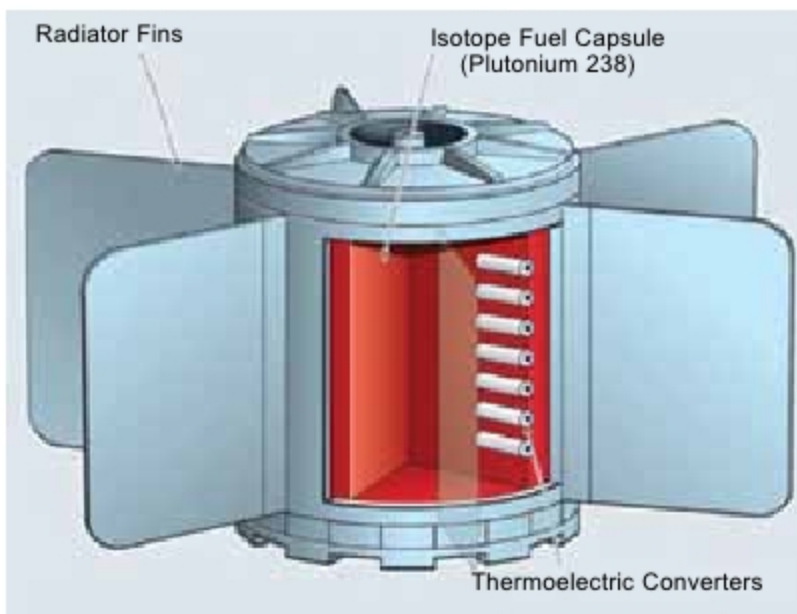
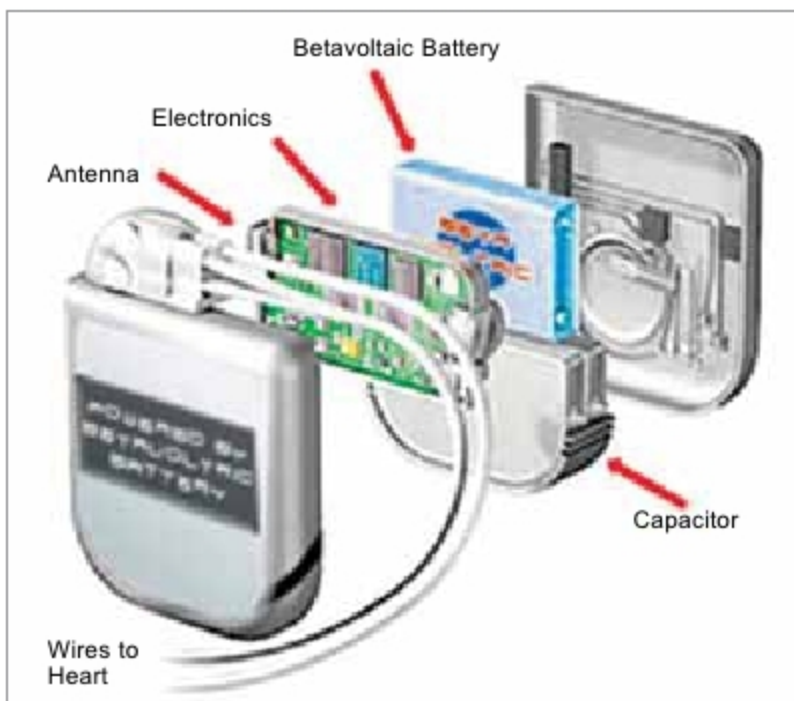




Main components of a radioisotope thermoelectric generator (RTG)
(Source: www.researchgate.net)



A betavoltaic battery powering a pacemaker (Source: www.altenergymag.com)

used in IMDs during the 1970s; however, the potential risk of radiation led to the shift towards lithium-ion batteries in the 1980s. With the recent advancements in nuclear batteries, the use of betavoltaic cells is being reconsidered.

Radioisotope thermal generators

Radioisotope thermal generators (RTGs) use thermal energy of a radioisotope decay to heat a thermocouple, which produces infrared radiation to power 'solar cells' by generating several hundred milliwatts of electrical power.

A large amount of radioactive material is required for an effective RTG, restricting its application because of high radiation dangers. NASA has been using RTG for the past thirty years or so. The United

States first launched a radioisotope thermoelectric generator (RTG) into space on a satellite in 1961. They have powered NASA's twin Voyager probes right from the year 1977 when these were launched. Having covered more than sixteen billion kilometres, the Voyagers are the most distant spacecraft from Earth and are still travelling fine.

RTGs have enabled many other missions, such as NASA's Mars Curiosity rover, to provide the communication links between the scientists and engineers on Earth and the missions in space. As RTGs are quite heavy, expensive, and inefficient, NASA is also experimenting with Stirling Engine patented by Robert Stirling in 1816.

The Stirling engine consists of two chambers—cold and hot. It uses a 'working fluid' (commonly air, helium, or hydrogen) with a regenerator or heat exchanger between the

two chambers. The difference in temperature and pressure between the two chambers causes the working fluid to expand and contract, resulting in a net conversion of thermal energy to mechanical energy.

Indirect conversion batteries

Indirect conversion batteries involve radioactive decay radiation consisting of alpha or beta particles impinging on radioluminescent material like phosphor to produce ultra-photons that are collected using photodiodes such as InAs, GaP, GaAs, and AlGaAs. The radioluminescent materials used are nanotubular or micro spherical structures filled with tritium thin films. The photons are guided to a remotely located photodiode. Waveguides are used to protect the diode from the radiation.

The overall efficiency of two per

cent at 3.5V is possible if the spectra are matched, and power conversion is optimal. The battery life depends on the half-life of the radioisotope used as well as on the rate at which the radioluminescent material degrades, which many times turns out to be faster than the rate at which the radioactivity falls.

Betavoltaic cells

Betavoltaic cells consist of a beta-emitting source, such as Tritium, coupled to a p-n diode.

Energetic beta particles (typically a few keV) emitted by the source hit the semiconductor, producing hundreds of thousands of electron-hole pairs. Most of them get swept across the electric field in the p-n diode to produce current. Some losses occur as some of the excited electron-hole pairs undergo recombination, producing photon and/or phonon.

Since betavoltaic cells can operate in extreme environmental conditions, these have replaced conventional chemical batteries as power sources in many low-power applications such as micro-electromechanical and electronic devices, implantable biomedical prosthetic devices, and military intelligence applications.

Betavoltaic cells have low-power output and are not suitable for consumer electronics where energy requirement is comparatively higher. A battery 1cm-squared wide and 0.2cm tall generates only one microwatt of power, whereas a smartphone needs about 1,857mW.

Application areas

Nuclear batteries have several advantages over the traditional sources of power, making them useful for applications in several fields. These radioisotope batteries have a high energy density and can function over wide ranges of environmental conditions of temperature, pressure, underwater, or in space.

The batteries have been tested in places as extreme as the top of a volcano and deep under the sea. Since radioisotope decay is sustained by the material itself, there is no need for refuelling or recharging.

The disadvantage of using these batteries is that their power density is lower than the chemical batteries,